

Propositions: Standard equivalencies

You may assume these and do not need to prove them in assessments unless asked to do so

- Idempotence (obvious from definition of operator)

$$P \wedge P \equiv P$$

$$P \vee P \equiv P$$

- Null elements (obvious from definition of operator)

$$P \wedge \text{False} \equiv \text{False}$$

$$P \vee \text{False} \equiv P$$

- Identity elements (obvious from definition of operator)

$$P \wedge \text{True} \equiv P$$

$$P \vee \text{True} \equiv \text{True}$$

- Contradiction and Excluded Middle (lecture 2)

$$P \wedge \neg P \equiv \text{False}$$

$$P \vee \neg P \equiv \text{True}$$

- Commutativity and associativity of conjunction and disjunction:

$$P \wedge Q \equiv Q \wedge P \quad (\text{both need to be true, so order is irrelevant})$$

$$P \vee Q \equiv Q \vee P \quad (\text{either needs to be true, so order is irrelevant})$$

$$(P \wedge Q) \wedge R \equiv P \wedge Q \wedge R \equiv P \wedge (Q \wedge R) \quad (\text{all need to be true})$$

$$(P \vee Q) \vee R \equiv P \vee Q \vee R \equiv P \vee (Q \vee R) \quad (\text{any need to be true})$$

- Distributivity of conjunction over disjunction, and vice versa:

$$P \wedge (Q \vee R) \equiv (P \wedge Q) \vee (P \wedge R) \quad (\text{check truth table if unsure})$$

$$P \vee (Q \wedge R) \equiv (P \vee Q) \wedge (P \vee R)$$

Consider for the first case: P must be true **and** either Q or R is true

Equivalently: either P **and** Q are both true **or** P **and** R must both be true

- Double negation (obvious from truth table and definition)

$$\neg\neg P \equiv P$$

- de Morgan (Seen in lecture2, remember that: $\wedge$ and $\vee$ swap)

$$\neg(P \wedge Q) \equiv \neg P \vee \neg Q$$

$$\neg(P \vee Q) \equiv \neg P \wedge \neg Q$$

- Definition of implication (seen last lecture)

$$P \Rightarrow Q \equiv \neg P \vee Q$$

- Deduced equivalences ('deduced' because you can work them out)

$$\text{True} \Rightarrow Q \equiv Q \quad (\text{Consider: True is P, } \neg\text{True} \vee Q)$$

$$\text{False} \Rightarrow Q \equiv \text{True} \quad (P \text{ is False, } \neg\text{False} \vee Q, \text{True} \vee Q)$$

$$P \Rightarrow \text{True} \equiv \text{True} \quad (Q \text{ is True, so this is } \neg P \vee \text{True})$$

$$P \Rightarrow \text{False} \equiv \neg P \quad (Q \text{ is False, so this is } \neg P \vee \text{False})$$

$$P \Rightarrow Q \equiv \neg Q \Rightarrow \neg P \quad (\text{next slide})$$

- More complex deduced equivalencies

$$P \Rightarrow (Q \Rightarrow R) \equiv (P \wedge Q) \Rightarrow R \quad (\text{see lecture 3})$$

$$P \Rightarrow (Q \vee R) \equiv (P \Rightarrow Q) \vee (P \Rightarrow R)$$

- Definition and rules for bi-implication:

$$P \Leftrightarrow Q \equiv (P \Rightarrow Q) \wedge (\neg P \Rightarrow \neg Q) \quad (\text{from definition})$$

$$P \Leftrightarrow Q \equiv Q \Leftrightarrow P \quad (\text{obvious from truth table})$$

$$P \Leftrightarrow \text{True} \equiv P \quad (\text{This is True iff P has value True})$$

$$P \Leftrightarrow \text{False} \equiv \neg P \quad (\text{This is True iff P has value False})$$